ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION â€" DECEMBER / JUNE 2024

FIRST YEAR (COMMON TO ALL BRANCHES) â€' SEMESTER 1

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted. Mobile phones are strictly prohibited.

--- UNIT I (Differential Calculus) ---

Q.1 (a) State and prove Rolle's Theorem. Verify it for f(x) = (x-1)(x-2)(x-3) in [1, 3]. [7 Marks]
(b) Expand e^(x) * cos(y) in powers of x and y using Maclaurin's Theorem up to 3rd degree terms. [7 Marks]

[OR]

Q.1 (Alt) Alternative: (a) If y = sin(m * sin^(-1) x), prove that (1 - x^2) y_(n+2) - (2n + 1)x y_(n+1) + (m^2 - n^2) y_n = 0 using Leibniz's Rule. [7 Marks]
(b) Find the radius of curvature at any point of the cycloid x = a(theta + sin theta), y = a(1 - cos theta). [7 Marks]

--- UNIT II (Partial Differentiation & Extrema) ---

Q.2 (a) If u = tan^(-1) [(x^3 + y^3) / (x - y)], prove that x(du/dx) + y(du/dy) = sin(2u) using Euler's theorem for homogeneous functions. [7 Marks]
(b) Find the maximum and minimum values of f(x, y) = x^3 + y^3 - 3axy. [7 Marks]

[OR]

Q.2 (Alt) Alternative: (a) Find the dimensions of a rectangular box open at the top of maximum capacity whose surface area is 432 sq. cm using Lagrange's Multipliers. [7 Marks]
(b) If x = r * sin(theta) * cos(phi), y = r * sin(theta) * sin(phi), z = r * cos(theta), evaluate Jacobian J = d(x,y,z) / d(r,theta,phi). [7 Marks]

--- UNIT III (Integral Calculus) ---

Q.3 (a) Evaluate the double integral ", (x^2 + y^2) dx dy over the region bounded by y = x^2 and y^2 = x. [7 Marks]
(b) Evaluate the triple integral "- x^2 y z dx dy dz over the volume bounded by x=0, y=0, z=0 and x+y+z=1. [7 Marks]

[OR]

Q.3 (Alt) Alternative: (a) Change the order of integration in "+[0 to 1] "+[x to " (2-x^2)] (x / " (x^2+y^2)) dy dx and evaluate. [7 Marks]
(b) Find the volume of the sphere x^2 + y^2 + z^2 = a^2 using spherical polar coordinates. [7 Marks]

--- UNIT IV (Linear Algebra & Matrices) ---

Q.4 (a) Find the rank of the matrix A by reducing it to Normal form (Echelon form). [7 Marks]
(b) Test for consistency and solve the system of linear equations: 2x - y + 3z = 8; -x + 2y + z = 4; 3x + y - 4z = 0. [7 Marks]

[OR]

Q.4 (Alt) Alternative: (a) State and prove Cayley-Hamilton Theorem. Verify it for matrix A = [[1, 2], [2, -1]] and find A^(-1) and A^4. [7 Marks]
(b) Diagonalize the symmetric matrix A = [[2, 2, -7], [2, 1, 2], [0, 1, -3]] by orthogonal transformation. [7 Marks]

--- UNIT V (Vector Calculus) ---

Q.5 (a) Find the directional derivative of f(x,y,z) = 2xy + z^2 at point (1, -1, 3) in the direction of vector i + 2j + 2k. [7 Marks]
(b) State Green's Theorem in a plane and verify it for ". [(xy + y^2) dx + x^2 dy] around the closed curve bounded by y = x and y = x^2. [7 Marks]

[OR]

Q.5 (Alt) Alternative: (a) Verify Stokes' Theorem for vector F = (2x - y)i - yz^2 j - y^2 z k over the upper half of sphere x^2 + y^2 + z^2 = 1. [7 Marks]
(b) State Gauss Divergence Theorem and evaluate the surface integral ", F · n dS over the cube bounded by x=0, x=1, y=0, y=1, z=0, z=1. [7 Marks]

**** END OF QUESTION PAPER â€" MITS GWALIOR AUTONOMOUS EXAMINATION SYSTEM ****